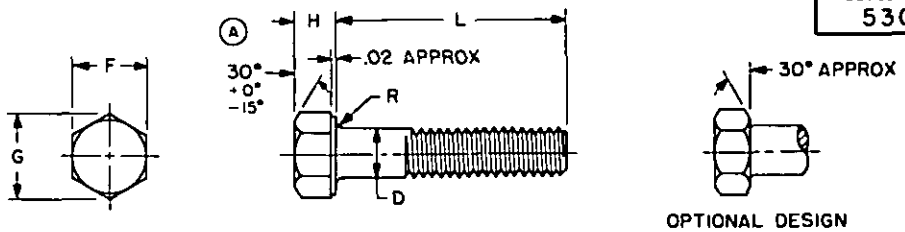
FED. SUP CLASS
5305



OPTIONAL DESIGN

NOMINAL SIZE		.250	.3125	.375	.4375	.500	.5625	.625
THREADS PER INCH (UNF-2A)		28	24	24	20	20	18	18
D BODY DIAMETER	MAX	.250	.3125	.375	.4375	.500	.5625	.625
	MIN	.245	.3065	.369	.4305	.493	.5545	.617
F WIDTH ACROSS FLATS	MAX	.4375	.500	.5625	.625	.750	.8125	.9375
	MIN	.4280	.489	.5510	.612	.736	.7980	.9220
G WIDTH ACROSS CORNERS	MAX	.505	.577	.650	.722	.866	.938	1.083
	MIN	.488	.557	.628	.698	.840	.910	1.051
H HEAD HEIGHT	MAX	.163	.211	.243	.291	.323	.371	.403
	MIN	.150	.195	.226	.272	.302	.348	.378
R RADIUS (A)	MAX	.025	.025	.025	.025	.025	.045	.045
	MIN	.015	.015	.015	.015	.015	.020	.020
TENSILE STRENGTH (LBS-MIN) (A)		2,910	4,640	7,020	9,490	12,490	16,240	20,480

L LENGTH	TOLERANCE (A) SIZE .375 AND UNDER	SIZES .4375 AND .500	SIZES .5625 AND .625	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
.375	+.000 −.030	+.000 −.030	+.000 −.030	1	27	53				
.438				2	28	54	79			
.500				3	29	55	80	104		
.562				4	30	56	81	105	130	
.625				5	31	57	82	106	131	155
.750				6	32	58	83	107	132	156
.875				7	33	59	84	108	133	157
1.000				8	34	60	85	109	134	158
1.125	+.000 −.040	+.000 −.060	+.000 −.080	9	35	61	86	110	135	159
1.250				10	36	62	87	111	136	160
1.375				11	37	63	88	112	137	161
1.500				12	38	64	89	113	138	162
1.750				13	39	65	90	114	139	163
2.000				14	40	66	91	115	140	164
2.250				15	41	67	92	116	141	165
2.500				16	42	68	93	117	142	166
2.750	+.000 −.060	+.000 −.080	+.000 −.100	17	43	69	94	118	143	167
3.000				18	44	70	95	119	144	168
3.250				19	45	71	96	120	145	169
3.500				20	46	72	97	121	146	170
3.750				21	47	73	98	122	147	171
4.000				22	48	74	99	123	148	172
4.250	+.000 −.100	+.000 −.100	+.000 −.100	23	49	75	100	124	149	173
4.500				24	50	76	101	125	150	174
4.750				25	51	77	102	126	151	175
5.000				26	52	78	103	127	152	176
5.500								128	153	177
6.000								129	154	178

(A) INACTIVE FOR NEW DESIGN AFTER 15 FEB 1973.

(A) For changes see sheets 1 and 2.

REVIEWER: AT, AV, IS, NS, OS, YD
USER: MC, ME, MI

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

APPROVED 4 MAR 1954 REVISED (A) 15 FEB 1973

P.A.	WC
Other Cust	SH 82

TITLE: SCREW, CAP, HEXAGON HEAD - NICKEL-COPPER ALLOY, PLAIN FINISH, UNF-2A

MILITARY STANDARD
MS 35312

PROCUREMENT SPECIFICATION: FF-S-85

SUPERSEDES:

SHEET 1 OF 2

DD FORM 672-1 (Continental)

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FED. SUP CLASS 5305

NOMINAL SIZE		.750	.875	1.000	1.125	1.250
THREADS PER INCH (UNF-2A)		16	14	12	12	12
D BODY DIAMETER	MAX	.750	.875	1.000	1.125	1.250
	MIN	.741	.866	.990	1.114	1.239
F WIDTH ACROSS FLATS	MAX	1.125	1.3125	1.500	1.6875	1.875
	MIN	1.100	1.2850	1.469	1.6310	1.812
G WIDTH ACROSS CORNERS	MAX	1.299	1.516	1.732	1.949	2.165
	MIN	1.254	1.465	1.675	1.859	2.066
H HEAD HEIGHT	MAX	.483	.563	.627	.718	.813
	MIN	.455	.531	.591	.658	.749
R RADIUS	MAX	.045	.065	.095	.095	.095
	MIN	.020	.040	.060	.060	.060
(A) TENSILE STRENGTH (LBS-MIN)		29,840	40,720	53,040	68,480	85,800

L LENGTH	TOLERANCE SIZE .750	TOLERANCE SIZES .875 AND 1.000	TOLERANCE SIZES 1.125 AND 1.250	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
.750	+.000	+.000	-----	179				
.875	-.030	-.040		180	202			
1.000				181	203	224		
1.125				182	204	225	245	
1.250				183	205	226	246	265
1.375				184	206	227	247	266
1.500	+.000	+.000	+.000	185	207	228	248	267
1.750	-.080	-.100	-.120	186	208	229	249	268
2.000				187	209	230	250	269
2.250				188	210	231	251	270
2.500				189	211	232	252	271
2.750				190	212	233	253	272
3.000				191	213	234	254	273
3.250	+.000	+.000	+.000	192	214	235	255	274
3.500	-.100	-.140	-.160	193	215	236	256	275
3.750				194	216	237	257	276
4.000				195	217	238	258	277
4.250				196	218	239	259	278
4.500				197	219	240	260	279
4.750	+.000	+.000	+.000	198	220	241	261	280
5.000	-.100	-.160	-.180	199	221	242	262	281
5.500				200	222	243	263	282
6.000				201	223	244	264	283

NOTES:

(A) 1. MATERIAL: Nickel-Copper alloy in accordance with QQ-N-281, Class A, having a minimum ultimate tensile strength of 80,000 PSI.

2. PROTECTIVE COATING: None.

3. THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services, Handbook H28.

4. DIMENSIONS: All dimensions are in inches, unless otherwise specified.

5. THREAD LENGTH: Minimum thread length shall be twice the diameter plus 0.250 inch. The tolerance shall be plus 0.188 inch or 2-1/2 threads, whichever is greater. On screws that are too short for minimum thread lengths, the distance from the head bearing surface to the first complete thread shall not exceed the length of 2-1/2 threads for sizes thru 1.000 and 3-1/2 threads for sizes over 1.000.

6. PART NUMBER: *The MS part number consists of the MS number, plus the dash number.* Example: MS35312-1.

7. Referenced documents shall be of the issue in effect on date of invitations for bid.

8. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

(A) INACTIVE FOR NEW DESIGN AFTER 15 FEB 1973.

REVIEWER:AT, AV, IS, NS
USER:MC, ME, MI, OS, YD

APPROVED 4 MAR 1954 REVISED (A) FOR CHANGES SEE SHEETS 1 AND 2.

This military standard is mandatory for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. WC	TITLE	MILITARY STANDARD
Other Cust SH 82	SCREW, CAP, HEXAGON HEAD – NICKEL-COPPER ALLOY, PLAIN FINISH, UNF-2A	MS 35312
PROCUREMENT SPECIFICATION FF-S-85	SUPERSEDES:	SHEET 2 OF 2

DD, FORM 672-1 (Continued) PREVIOUS EDITION OF THIS FORM ARE OBSOLETE.

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